

Nathanael Rosenheim, Southeast Texas Urban Integrated Field Lab Demographics for Community Resilience Planning

People First Models

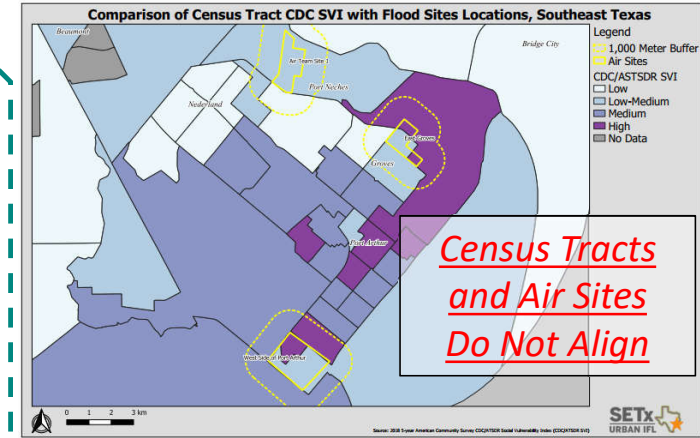


The Subject

To understand disparities in resilience, researchers must combine population, hazard, and impact data.
Data integration is not easy.

The Approach

- Census data provides the baseline for population data within complicated geographies.
- Our approach disaggregates and transforms the data into housing units linked to individual structures.



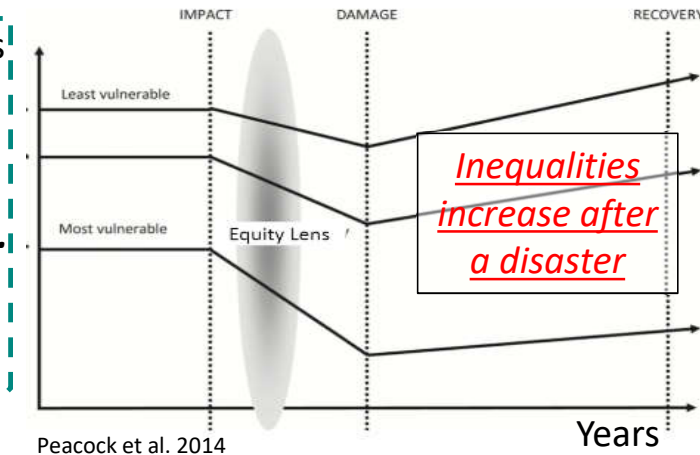
Census Tracts and Air Sites Do Not Align

Simplified data integration allows for the detailed exploration of how people are impacted by any mappable hazard, risk, and building level impacts.

The Outcomes

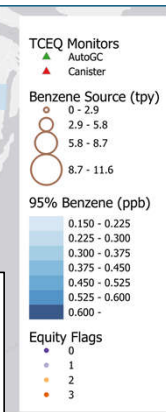
Underserved communities experience greater hazard burdens and slower recovery.
Change requires evidence. Evidence requires good data integration.

Why Care?



Inequalities increase after a disaster

Air Pollution + Household Equity

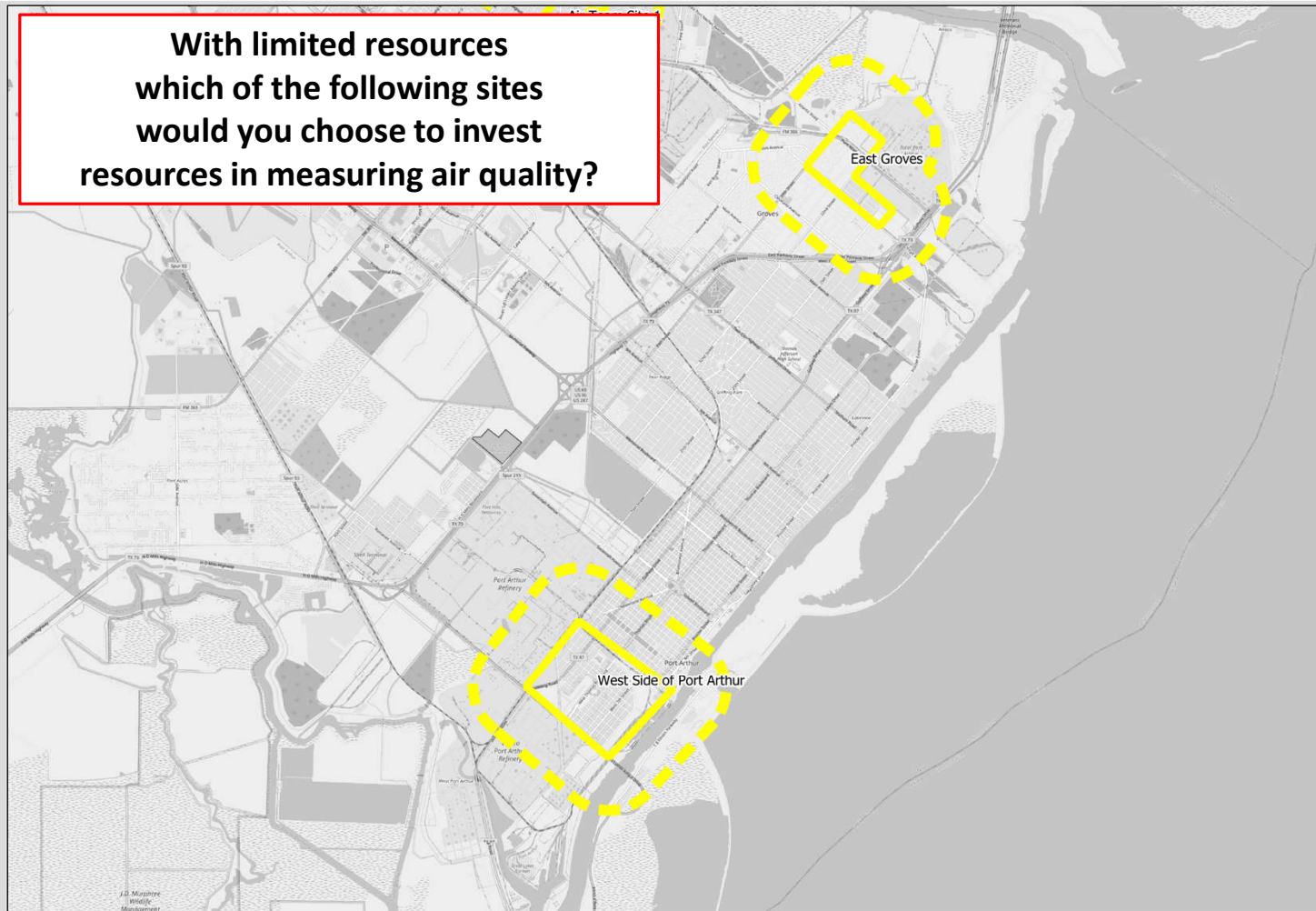


Potential Air Sites within Port Arthur in Southeast, TX

**With limited resources
which of the following sites
would you choose to invest
resources in measuring air quality?**

Legend

-  1,000 Meter Buffer
-  Air Sites



0 1 2 3 km



Comparison of Census Tract CDC SVI with Air Site West Side of Port Arthur in Southeast, TX



Comparison of Census Tract CDC SVI with Air Site East Groves in Southeast, TX



Provenance: fu3m12_3gr2_3intersectSitesMUA_2024-04-11.qgz

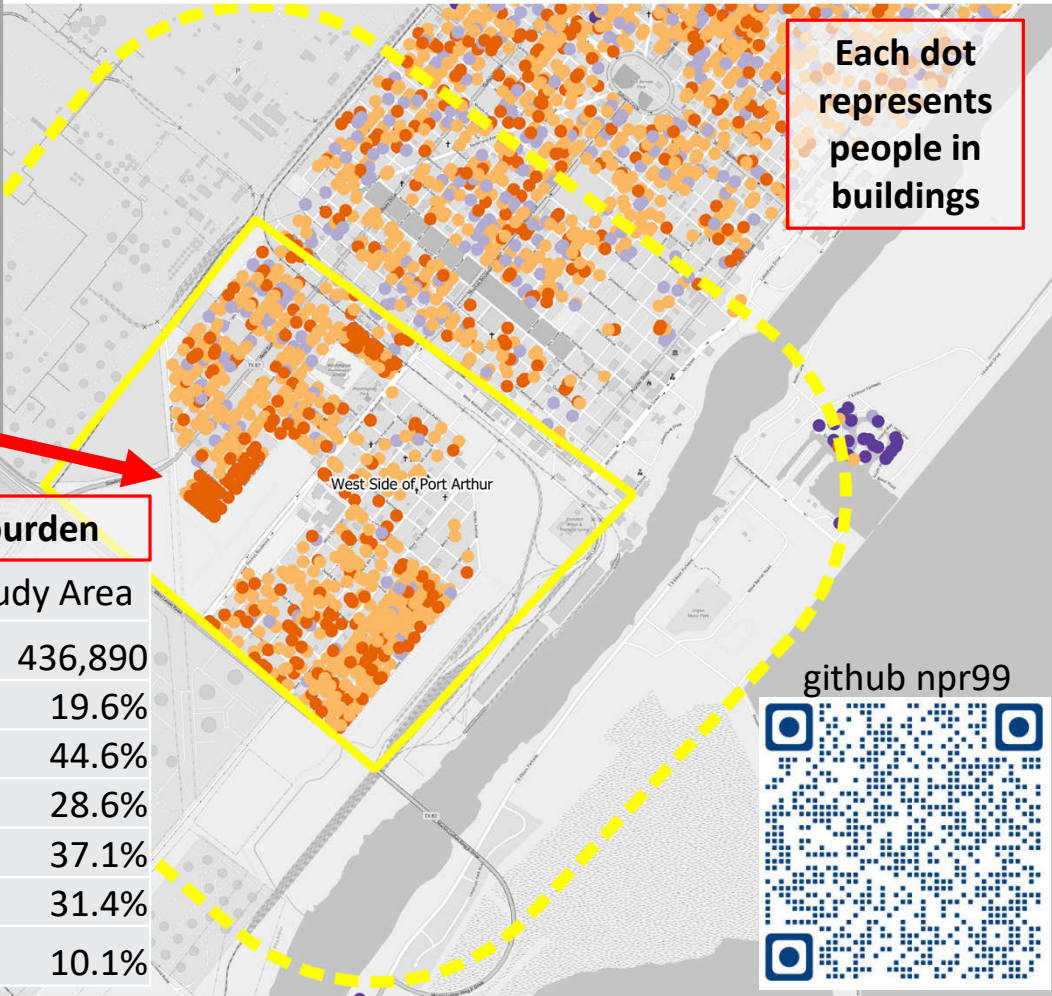
Comparison of Household Level Equity Flags with Air Site West Side of Port Arthur in Southeast, TX



Kids playing basketball just next to oil refineries in Port Arthur (Photo: Hilton Kelley)

Evidence of disproportionate burden

	Air Site	Study Area
Total Population	4,096	436,890
Poverty (%)	43.6%	19.6%
Low Income (%)	74.3%	44.6%
Renter (%)	52.4%	28.6%
Minority (%)	97.7%	37.1%
0 Equity Flags (%)	0.3%	31.4%
3 Equity Flags (%)	37.5%	10.1%



Legend

1,000 Meter Buffer

Air Sites

Equity Flags

0

1

2

3

1 dot = 1 household in a building

Equity Flags Include:

- Income below median (\$45,000)

- Renter

- Minority

0 Equity flags represent a White non-Hispanic homeowner with income greater than \$45,000

1 Equity flag represents a household that either has low income, is a renter, or a minority.

2 Equity flags represent some combination of 2 flags. For example, a low income renter.

3 Equity flags represent a household that has an income below the median, is a renter and a minority.

Comparison of Household Level Equity Flags with Air Site East Groves in Southeast, TX

